Oscillation: Measuring the Local Irregularity of Functions

This document was generated by a large language model (LLM) and may contain errors.

Contents

1 Introduction

In the previous chapter we studied continuity and the various ways in which it can fail. We learned that discontinuities come in several forms—removable, jump, infinite, and oscillatory. While this classification is useful, it is qualitative. It tells us *what kind* of discontinuity is present but does not measure *how irregular* a function is near a point.

The concept of *oscillation* provides exactly such a measurement.

Instead of asking whether a function is continuous, oscillation asks a more refined question:

How much can the function vary inside an arbitrarily small neighborhood of a point?

Remarkably, this simple idea connects several major topics in real analysis.

- Continuity can be characterized by zero oscillation.
- Jump discontinuities correspond to positive oscillation.
- Highly oscillatory functions exhibit maximal oscillation.
- Riemann integrability can be described entirely in terms of oscillation.

Oscillation therefore serves as a bridge between continuity and integration.

2 Variation on a Set

Before discussing oscillation at a single point, it is useful to measure how much a function varies on an arbitrary set.

Definition 2.1. Let $E \subseteq \mathbb{R}$ and suppose that f is bounded on E.

The oscillation of f on E is

$$\operatorname{osc}(f, E) = \sup_{x,y \in E} |f(x) - f(y)|.$$

This quantity measures the largest possible difference between function values inside the set.

Equivalently,

$$\operatorname{osc}(f, E) = \sup_{x \in E} f(x) - \inf_{x \in E} f(x).$$

The second formula is often more convenient in analysis.

3 Examples

Example 3.1. If

$$f(x) = 5,$$

then

$$\mathrm{osc}(f, E) = 0$$

for every set E.

The function never changes value.

Example 3.2. Let

$$f(x) = x$$

on

$$E = [1, 4].$$

Then

$$\mathrm{osc}(f, E) = 4 - 1 = 3.$$

Example 3.3. For

$$f(x) = \sin x$$

on

$$E = [0, 2\pi],$$

we obtain

$$\mathrm{osc}(f, E) = 2,$$

since the function ranges from -1 to 1.

4 Local Oscillation

The previous definition measures global variation over a set.

To understand continuity we must instead measure variation inside progressively smaller neighborhoods.

For $\delta > 0$, define

$$I_\delta(x) = (x - \delta, x + \delta).$$

The oscillation of f inside this neighborhood is

$$\omega_f(x, \delta) = \sup_{u,v \in I_\delta(x)} |f(u) - f(v)|.$$

As δ decreases, the neighborhood shrinks.

Global behavior gradually disappears, leaving only the local behavior near the point.

5 Oscillation at a Point

We are now ready for the central definition.

Definition 5.1. Suppose f is bounded in some neighborhood of x.

The oscillation of f at x is

$$\omega_f(x) = \lim_{\delta \to 0} \omega_f(x, \delta).$$

Why does this limit exist?

Observe that if

$$0 < \delta_1 < \delta_2,$$

then

$$I_{\delta_1}(x) \subseteq I_{\delta_2}(x).$$

Therefore,

$$\omega_f(x, \delta_1) \leq \omega_f(x, \delta_2).$$

Thus the oscillation cannot increase as the neighborhood becomes smaller.

The function

$$\delta \longmapsto \omega_f(x, \delta)$$

is monotone decreasing.

Since it is also bounded below by zero, the limit always exists (possibly equal to zero).

6 Interpretation

Oscillation provides a numerical measure of local irregularity.

If

$$\omega_f(x) = 0,$$

then every sufficiently small neighborhood contains function values that are arbitrarily close together.

The function becomes nearly constant when viewed under sufficient magnification.

If

$$\omega_f(x)$$

is positive, then no matter how closely we zoom in, there remain points whose function values differ by at least some fixed amount.

Thus oscillation measures how stubbornly the function refuses to settle down.

7 Examples

7.1 Continuous Functions

Every continuous function satisfies

$$\omega_f(x) = 0$$

at every point.

Indeed, continuity states that nearby function values become arbitrarily close to the function value at the center point.

7.2 Jump Discontinuities

Consider

$$f(x) = \begin{cases} 0, & x < 0, \\ 1, & x \geq 0. \end{cases}$$

Every neighborhood of the origin contains points whose values are 0 and 1.

Hence

$$\omega_f(0) = 1.$$

Away from the origin,

$$\omega_f(x) = 0.$$

7.3 The Floor Function

The floor function

$$\lfloor x \rfloor$$

has oscillation

$$1$$

at every integer.

Elsewhere,

$$\omega_f(x) = 0.$$

7.4 A Removable Discontinuity

Consider

$$f(x) = \begin{cases} 1, & x = 0, \\ 0, & x \neq 0. \end{cases}$$

Every neighborhood of the origin contains both values 0 and 1.

Consequently,

$$\omega_f(0) = 1.$$

Although the discontinuity is removable, its oscillation is still positive.

7.5 Rapid Oscillations

Now consider

$$f(x) = \sin\left(\frac{1}{x}\right), \qquad x \neq 0.$$

Every neighborhood of the origin contains points where the function equals 1 and points where it equals -1.

Therefore

$$\omega_f(0) = 2.$$

This is the largest possible oscillation for a function whose values lie between -1 and 1.

7.6 Dirichlet's Function

Define

$$f(x) = \begin{cases} 1, & x \in \mathbb{Q}, \\ 0, & x \notin \mathbb{Q}. \end{cases}$$

Every interval contains both rational and irrational numbers.

Hence

$$\omega_f(x) = 1$$

for every real number.

The function is therefore discontinuous everywhere.

7.7 Thomae's Function

Thomae's function is defined by

$$f(x) = \begin{cases} 1/q, & x = p/q \text{ in lowest terms}, \\ 0, & x \notin \mathbb{Q}. \end{cases}$$

This remarkable example satisfies

$$\omega_f(x) = 0$$

at every irrational point,

but

$$\omega_f(x) > 0$$

at every rational point.

It is continuous exactly at the irrational numbers.

8 Characterizing Continuity

Oscillation provides one of the cleanest characterizations of continuity.

Theorem 8.1. *A function is continuous at a point x if and only if*

$$\omega_f(x) = 0.$$

Thus continuity is equivalent to saying that the function becomes arbitrarily flat when viewed under sufficiently high magnification.

9 Oscillation and Upper and Lower Sums

Oscillation plays a central role in Riemann integration.

Consider a subinterval

$$I = [a, b].$$

The oscillation of f on I is

$$\sup_I f - \inf_I f.$$

If the oscillation is small, then every value of the function on that interval is nearly the same.

Consequently,

- the upper rectangle,
- the lower rectangle,

have nearly equal heights.

Their contribution to the difference between the upper and lower Riemann sums is therefore small.

Large oscillation produces large discrepancies between upper and lower sums.

10 Oscillation and Riemann Integrability

The importance of oscillation culminates in one of the central theorems of real analysis.

Theorem 10.1 (Lebesgue's Criterion for Riemann Integrability)**.** *A bounded function on a closed interval is Riemann integrable if and only if the set of points where*

$$\omega_f(x) > 0$$

has measure zero.

Although measure theory lies beyond the scope of this chapter, the theorem has a simple interpretation.

A bounded function is Riemann integrable provided its discontinuities occupy a negligibly small portion of the interval.

This explains why piecewise continuous functions are always Riemann integrable.

11 Oscillation Versus Other Measures of Regularity

Oscillation measures only local variation.

Other notions of regularity describe different aspects of a function.

Continuity	Local predictability
Uniform continuity	Global control of continuity
Differentiability	Local linear approximation
Lipschitz continuity	Bounded rate of change
Bounded variation	Total accumulated variation
Oscillation	Local spread of function values

These concepts complement one another rather than compete.

12 A Geometric View

Imagine repeatedly zooming into the graph of a function.

For a continuous function, every magnification reveals a graph that becomes flatter and flatter.

The oscillation tends toward zero.

For a jump discontinuity, every zoom still reveals two separated branches.

The oscillation never disappears.

For

$$\sin\left(\frac{1}{x}\right),$$

every magnification uncovers new oscillations.

No matter how closely we inspect the graph, it never resembles a continuous curve.

Oscillation therefore captures exactly what our geometric intuition observes.

13 Summary

Oscillation transforms continuity from a binary concept into a quantitative one.

Instead of merely asking whether a function is continuous, it measures how much the function continues to vary inside arbitrarily small neighborhoods.

This single idea unifies several seemingly unrelated topics.

It explains why continuous functions have stable local behavior, why jump discontinuities remain visible under magnification, why rapidly oscillating functions resist limiting values, and why Riemann integration depends on the distribution of discontinuities rather than their mere existence.

For these reasons, oscillation occupies a central position in real analysis, providing one of the most elegant connections between continuity, integration, and the local geometry of functions.